Effect of Feedback Styles in Robotic Fitness Coaching on Exercise Intensity

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Abstract Intensive exercise effectively prevents adult diseases, such as cardiovascular disease; however, many people find it challenging to keep this routine due to many situational and financial constraints. Recently, a growing interest has been in deploying social robots as fitness coaches in exercise sessions. This study investigated the effectiveness of real-time feedback provided by robot coaches in increasing exercise intensity. Participants were randomly assigned to either an empathic feedback group, focusing on emotional support, or a factual feedback group, offering information and advice only. All participants engaged in physical exercises (e.g., stretching and treadmill walking), receiving feedback on their walking speed and posture at 20-second intervals for 10 to 30 minutes. By analyzing variations in heart rate and treadmill speed, the study assessed the impact of the robot coach's intervention on changes in exercise intensity. The results indicated that participants receiving factual feedback experienced significant heart rate and walking speed increases, maintaining higher walking speeds for extended periods. However, the post-experiment interview results revealed that while many participants preferred factual feedback, they also highlighted the importance of the robot's emotional support (e.g., encouragement) in enhancing the overall exercise experience. In conclusion, the current study suggests that while informative feedback effectively improves actual exercise intensity, emotional support from a robot coach is also crucial for fostering long-term effects through continuous interaction.

핵심어: Socially assistive robot (SAR), Robot coaching, Exercise feedback, Cardiovascular exercise, Empathy

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1. Introduction

The growing trend of increased indoor dwelling and limited physical activity intensifies global health concerns, particularly regarding cardiovascular diseases. According to a 2022 report from the World Health Organization, about 25% of adults and over 80% of adolescents lead an inactive lifestyle, resulting in severe health repercussions[1]. This surge in sedentary behavior worldwide demands accessible and innovative fitness solutions that promote regular physical activity.

Previous research has focused on enhancing exercise motivation through traditional methods like individualized in-person exercises or group activities[2,3]. Collaboration with robots has gained interest in making physical exercise more engaging. Robots serve effectively as fitness companions, fostering social bonds with users when designed for appropriate interaction[4,5]. For example, in an exercise program designed to help beginners complete a 5 km running course, participants who received personalized robotic coaching demonstrated higher motivation to continue and improve their performance[6].

The impact of feedback provided by social robots on the actual physiological and behavioral changes in exercise intensity remains underexplored. For example, research has focused on measuring user perceptions of these robots by older adults[7] or the feedback effect on stroke rehabilitation[8,9].

Therefore, it is necessary to investigate how different forms of encouragement from robotic coaches affect exercise behaviors, drawing on physical metrics such as heart rate and walking speed. This study investigates how feedback emphasizing factual information, as opposed to emotional support, influences the efficacy of robot-assisted physical exercise.

2. Related Work

2.1 Effect of Social Presence on Human Performance

The presence and feedback of professional coaches play a crucial role in creating engaging exercise experiences. Research indicates that the immediacy and quality of feedback from coaches can significantly influence exercise intensity and motivation, helping maintain appropriate exercise levels[10], while peer influence may extend exercise duration and increase enjoyment[11]. The impact of a coach, however, varies depending on the coach-athlete relationship, feedback style, the athlete's personality, and the exercise's nature[12].

Social Facilitation Theory suggests that the mere presence of others can enhance individual performance[13]. This is particularly evident in group exercise settings, where the awareness of others can lead to heightened performance[14].

The influence of social presence extends beyond human interactions, as the presence of robots can also affect users' motivation, comparable to the impact of human counterparts[15-17]. This phenomenon is explained by the 'Computer as Social Actors (CASA)' paradigm[18], which posits that individuals often extend the same social norms, expectations, and attitudes to robots that display human-like attributes, including appearance, vocal expression, and gestures[19,20].

2.2 Robot as Fitness Coach

Although human guidance is beneficial in physical exercise routines, not everyone has access to personalized, in-person support. Factors contributing to this include the rise in single-person households, economic disparities, and an aging population, making it imperative to develop accessible health interventions tailored to diverse living situations and user needs.

One promising solution is robot-assisted intervention with social robots or, more specifically, robots that supplement the efforts of human professionals. These specialized robots, called *Socially Assistive Robots* (SARs), are designed to cater to human needs through interactive services, functioning as companions or personal assistants[21,22]. This companionship can help people maintain a more independent and healthy lifestyle (e.g., aging-in-place)[23].

Past studies within human-robot interaction have found SARs' effectiveness in increasing engagement in the program, which is a critical step for building healthy habits. For instance, a robot developed by Kidd and Breazeal has been utilized in weight loss programs, serving as an effective guide[21]. Robot coaching is effective for all age groups, for both younger[22] and older adults[23,24], in making the exercise more engaging. SARs with physical embodiment, in particular, like Paro, have shown an advantage in inducing more social engagement among users over virtual agents, even long after the termination of the experiment[25], leading to increased user compliance[26] and task performance[27].

While SARs have been influential in guiding and enhancing motivation for physical activity, much of the existing research has not concentrated on their impact on exercises specifically related to cardiovascular health, such as walking, and its effect on heart rates. Walking speed and heart rate are pivotal in determining the effectiveness of exercise. Specifically, maintaining a brisk walking pace and an appropriately elevated heart rate can amplify the advantages of walking, improving cardiovascular health[28], aiding weight control[29], enhancing overall well-being[30], and even diminishing the risk of premature mortality[31].

2.3 Role of Robots' Empathy Expression in Exercise Intensity

Empathy involves understanding and sharing others' emotions[32,33], which is essential for building social connections and promoting well-being.

The system's ability to understand user emotions plays a central role in forming meaningful connections, immersing users in the experience with virtual avatars, and ultimately boosting engagement between humans and robots[19,34]

The capacity of a robot to use empathic expression is vital, especially for establishing bonds that mirror human-human interactions. Research has shown that various forms of empathetic expression by robots significantly influence user attitudes towards both physical and virtual social robots[25,35-40]

Robots' emotional support includes calling users' names, praising users to motivate for a provided task, and showing concern following failed attempts[38,39] For example, in the Bickmore and Picard[38] study, robots with these empathic abilities, referred to as relational agents, were found to be more engaging and respected by users than those not equipped with these social skills, even four weeks after the termination of the experiment.

The impact of varied feedback types on physical exercise is yet under-explored, particularly in increasing walking speed and heart rate. For instance, a recent study by Irfan et al.[20] employed diverse feedback strategies to maintain the motivation of patients undergoing cardiac rehabilitation sessions, focusing on walking exercises. This feedback primarily consisted of continuous encouragement from the beginning through to the cool-down phase of the training, with prompts like, "You can do it!" and "Only a few minutes left!" Additionally, researchers offered factual instructions regarding the patients' cervical posture, such as advising them to look straight ahead if they were looking down and heartbeat monitoring. Despite this, a rigorous investigation of the effects of such feedback, mainly when differentiated by emotional support or factual information, is necessary.

2.4 The Present Study

Although having a coach around during exercise can boost intensity and motivation since its impact is not always straightforward, we must explore sub-factors supporting the practical exercise. We aim to elucidate the potential of robotic coaches in inducing more walking activity through different forms of coaching feedback.

R1: Does real-time feedback from robots effectively encourage people to exercise?

R2: How do different styles of feedback by robotic coaches influence individuals ' exercise intensity?

3. Methods

3.1. Research Design

We used a between-subjects experimental design to examine the effects of different types of robotic feedback-factual and empathic-on exercise intensity and user experience. Participants were randomly assigned to one of two groups: Group 1 received factual feedback, and Group 2 received empathic feedback from the robot coach. We measured behavioral measures (walking time and speed), physiological measures (heart rate), and perceived usability and effectiveness. This work was approved by our institution's IRB.

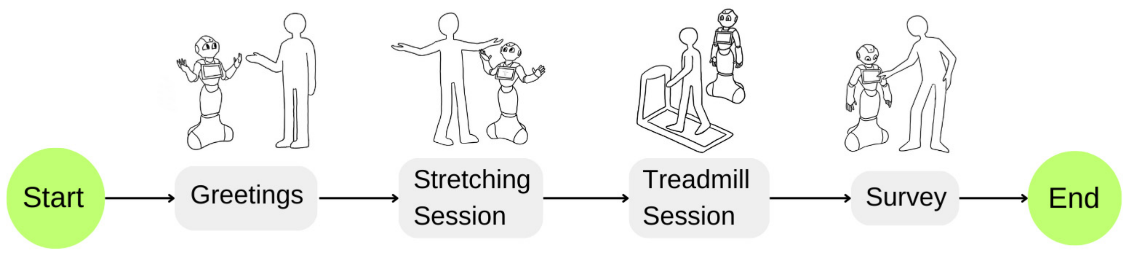

Figure 1. Overall experiment procedure

3.2 Participants

A total of 45 healthy adult participants (25 females) without musculoskeletal disorders were recruited. Participants were randomly assigned to one of two groups: Group 1 (factual feedback), with 22 participants, and Group 2 (empathic feedback), with 23 participants. All participants provided informed consent.

3.3 Robot Setup

The robot used in this study was Pepper (model: Pepper 1.8), a humanoid model designed by SoftBank Robotics. Pepper stands at a height of 1.2 meters and is equipped with a 3D camera and four microphones in its head to perceive its surroundings. Its design incorporates three omnidirectional wheels for mobility, and it can perform a range of gestures with its articulated hands, arms, and torso. Pepper was programmed to listen to human participants' speech, obtain input via its equipped touchscreen and a tablet PC, mimic exercise poses, and provide verbal feedback, enabling it to serve as an interactive exercise robot coach. We selected a humanoid robot specifically for its human-like features, such as making eye contact and demonstrating active listening behaviors like turning its head during communication. These characteristics and physical presence were expected to encourage participants' motivation to continue exercising.

The robot operated on the Android OS. Communication with Pepper was established through TCP/IP socket communication, allowing for real-time transmission of data and commands. This setup was facilitated through an app running on Android OS, developed using the Java programming language in Android Studio. We created a custom protocol including port numbers and data types necessary for socket communication.

Two main applications were involved: a server app on a tablet PC monitored by the researcher, and a client app on Pepper. As the robot progressed through each stage within sessions, the client app sent status updates to the server app via socket communication. Data received from the client was stored in the server app's files. Survey results displayed on the server app were stored in JSON format, then parsed and converted to TXT files for easier access and analysis.

3.4 Procedure

3.4.1. Initial Interaction

Each session began with a light ice-breaking talk where Pepper greeted participants and checked their mood. Based on the participants' mood (positive, neutral, negative), Pepper responded accordingly to make them comfortable. This was aimed at easing any participant who may have encountered a humanoid robot for the first time.

3.4.2. Stretching Session

Participants excercised a 3.5-minute stretching session following a tutorial video on a tablet PC(Figure 2a). Pepper mimicked the exercise poses or performed cheering motions if the poses were beyond its range of motion or too fast to catch up. The activity consisted of different stretching postures stimulating various body parts for a full-body stretch before the walking session.

3.4.3. Walking Session

Participants then walked on a treadmill placed beside Pepper (Figure 2b). The initial speed level of the treadmill was set to 0 km/hr ('stop' mode), and participants could adjust the walking speed ranging from 0 to 8 km/hr. Pepper provided feedback every 30 seconds, including cheering, speed suggestions, and posture suggestions. Throughout the walking session, participants could freely adjust the treadmill's speed. An observation camera,

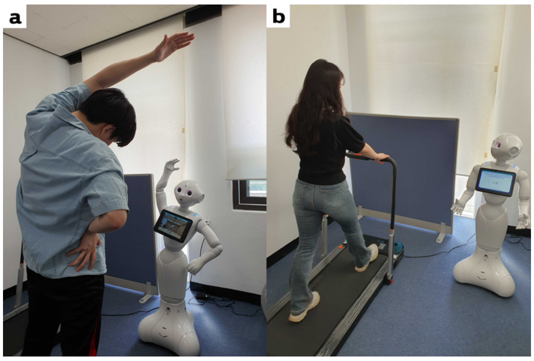

Figure 2. Treadmill and the robot coach setup

iPhone 13, recorded the change in speed. Heart rate was measured using the 'Health' app on an Apple Watch Series 6 (GPS) paired with an iPhone 13. An observation camera recorded changes in treadmill speed and the angle was set to record the treadmill's speed display.

During the treadmill walking session, the robot coach provided either one of two types of feedback: 1) Factual Feedback (FF): Group 1 received precise postural advice (e.g., "Pull your chin back, open your shoulders, and walk straight"). 2) Empathic Feedback (EF): Group 2 received emotional support with minimal factual information (e.g., "You are doing great. Pull your chin back, open your shoulders, and walk straight").

Each condition had one common feedback option to encourage walking and four different feedback options (e.g., in-toeing gait, out-toeing gait, forward head posture, encouragement), randomly provided to participants. Both feedback types were implemented with the same frequency, delivering feedback every 20 seconds but varied in content based on their respective emphasis.

3.4.4. Survey

After the walking session, participants completed an online survey that was displayed on the tablet PC equipted on robot's torso, and participants answered the survey by clicking the buttons. We collected the survey data through a tablet PC equipped on the robot coach. Each item had the form of a 5-point Likert scale. To build the questionnaire, we referred to previous research on human-robot interaction.

We mainly asked three subscales: *acceptance, perceived usefulness*, and *robot appearance*. Some of the items for reverse coded - lower the score, the higher level of each subscale- to check validity (and marked with '*' in this paper). The full version of the questionnaire is available in Appendix I.

For *acceptance* and *perceived usefulness,* we referred to questions from previous studies that used robot coaching in similar contexts[7,41-43]. Sackl et al.[7] examined acceptance and the persuasiveness of robot instruction quality, with original questions such as "I enjoyed the training with the robot very much" and "The robot supported and motivated me to do the exercise correctly." Fasola and Mataric[41] evaluated the quality of repetitive verbal instructions and perceptions of the robot's competence and overall helpfulness. G rer et al.[42] and Čaić et al.[43] focused on perceived warmth, competence, and automated social presence, including questions like "I feel the human/robotic coach understands me" and "I think the human/robotic coach is competent."

For *robot appearance*, we referred to studies such as Haring et al.[44], which examined the impact of physical embodiment on human compliance. The original study included questions about human likeness and compliance, which we adapted as follows: "The robot moves too fast for me to follow," "The robot has a clear voice," "I didn't quite understand the robot's instructions," and "I think the size of the robot is appropriate for exercise."

3.4.5. Post-experiment Interview

Prior to the exercise session, participants were informed and asked to wait until the researcher returned after finishing the survey. Once the survey was complete and submitted, the researcher received an automatic alarm and returned to the experiment room to continue a post-experiment interview. The researcher, who was waiting outside, then returned to conduct the post-experiment interview. We asked them to freely share their thoughts and the overall experience. The interview took about 15 minutes on average.

4. Results

4.1 Behavioral Changes

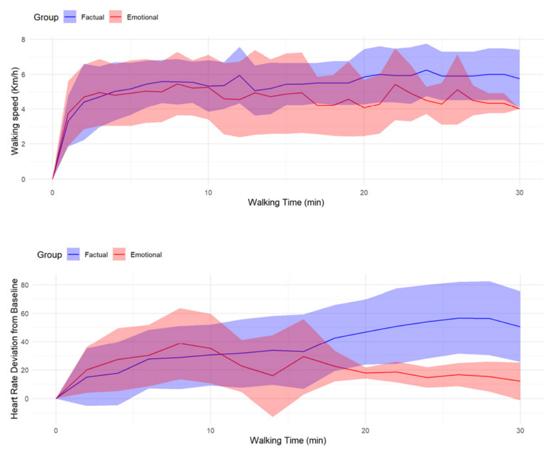

Figure 3. Average heart rate change (Top) and walking speed change (Bottom) by treadmill walking time

A study compared two groups using robotic coaching; FF (Group 1) and EF (Group 2) respectively. The study analyzed the total walking times, changes in walking speed, and changes in heart rate. All participants engaged in a single walking session with no interruptions, ranging from 10 min and 13 s to 33 min and 26 s. While Group 2 had a slightly longer walking time

The average walking time for the FF group was 980.64 s (*SD* = 495.78) which is 16 min and 20 s while the EF group was 983.22 s. (*SD* = 483.54) which is 16 min and 23 s. No significant difference was observed between the two groups ($t(42)$ = −0.018, p = 0.986).

The change in walking speed was collected during the walking session and was analyzed by manual coding of the video footage of the treadmill screen recorded during the session. The data was recorded in minutes and averaged across groups. Figure 3 shows the change in walking speed (top figure) over time for two groups. Significant differences were observed between the two groups from one minute after initiation, the required time to adjust the initial pace, until the termination of walking.

Group 1 demonstrated a higher level of walking speed ($M_{walkingspeed}$= 4.95 km/h, *SD* = 1.97) compared to Group 2 ($M_{walkingspeed}$ = 4.51km/h, *SD* = 2.11), and this difference was statistically significant ($t(758)$ = 2.968, p = .003). Therefore, Group 1 maintained a higher walking speed throughout the duration than Group 2.

We collected data on the heart rates of each participant in 2-minute intervals using a smartwatch. We analyzed the maximum heart rate recorded during each interval. In line with the change in walking speeds, a significant difference was also identified between the two groups regarding heart rate increase. We excluded missing data. As seen in Figure 3 (bottom), Group 1 exhibited a significantly higher rise in heart rate ($M_{heartrate}$= 30.13 beats/min, *SD* = 25.87, *N* = 16) compared to Group 2 ($M_{heartrate}$= 23.02 beats/min, *SD* = 21.61, *N*= 15). This observed difference in heart rate increment was statistically significant (p = .01), demonstrating that Group 1 experienced a greater elevation in heart rate during the period.

4.2 Survey

4.2.1 Acceptance

Participants showed a high preference for using the robot as a fitness coach (*M* = 3.58, *SD* = 0.92) and were willing to reuse the robot coach for future sessions (*M* = 3.87, *SD* = 0.92). The smartwatch that monitored heartrate did not intervene in the process as it showed minimal discomfort (*M* = 4.4, *SD* = 0.78). Participants were satisfied with the robot' s expression of emotional support (*M* = 3.42, *SD* = 0.81)

4.2.2 Perceived Usefulness

The robot coach's feedback was expected to be highly effective in helping participants maintain a healthy routine (*M* = 4.27, *SD* = 0.62). The movements of the robot were perceived as accurate (*M* = 3.09, *SD* = 0.82) and highly motivated participants to continue exercise (*M* = 4.00, *SD* = 0.80). Participants found the robot moderately helpful (*M* = 4.11, *SD* = 0.68), easy to use (*M* = 4.07, *SD* = 0.78), and trustworthy (M = 3.87, *SD* = 0.79).

4.2.3 Robot Appearance

The speed of robot movement was not too fast to follow (*M* = 4.83, *SD* = 0.72). Audio sound (*M* = 3.56, *SD* = 1.01) and instructions were clear (*M* = 4.09, *SD* = 0.82). Participants rated the robot's physical appearance and size as moderately appropriate as a fitness assistant (*M* = 3.6, *SD* = 0.84)

In short, the result showed that participants were willing to reuse the robot for future exercise, and perceived the robot coach as helpful to maintain a healthy routine. Participants did not feel the usage of the smartwatch was an obstacle to focus on the exercise. Also, participants did not feel the movement of the robot coach too fast. No significant difference was observed between Group 1 and Group 2 from this survey.

4.3 Post-experiment Interview

After filling out the questionnaire, participants were invited to interview and share their thoughts on the experience freely. At the end of the interview, participants evaluated the overall satisfaction level of the exercise experience they had with the robot coach on a scale of 5 points. The score showed that they had a positive experience with the robot coach (M = 4.36, SD = 0.55). We found several interesting findings via post-experiment interview and here is the qualitative result analysis.

First, participants' daily exercise habits led to different behavioral change in exercise intensity. Participants who regularly exercise already reported that they did not feel the need to alter their walking speed or any exercise intensity significantly, and especially just because the robot told them to do so. This means that when designing the future exercise sessions for people with different exercise routine profile, the exercise sessions should be longer than 30 minutes and speed limit should be higher than 8, because they report that they prefer exercise at a high speed for longer time. People who regularly exercise also more inclined to listen to the information provided by the robot, paying more attention to the posture which may have affected the change in walking speed. For example, participant #27 commented "I didn't exercise more than usual in this session because if I walk too fast it would be harder to pay attention to what my posture looks like." In contrast, participants who do not regularly exercise or not fond of working out on a treadmill showed a notable increase in exercise intensity. The robot's presence contributed significantly to making the exercise session more enjoyable. Participants mentioned that having the robot alongside them added an element of fun to the routine. For example, Participant #11 stated, "I don't usually exercise much, but with the coaching robot, I exercised more intensely."

The spontaneity of feedback provided in real time had a positive influence on overall participant satisfaction. Participants stated that they appreciated not feeling alone during the exercise sessions and valued the immediate guidance provided by the robot. In addition, many participants felt that exercising with the robot offered a flexible and constraint-free alternative to human-only guided professional personal training, due to fewer spatial, temporal, and financial limitations.

However, participants stated several challenges and considerations for more efficient and long-term effect. First, the main issue derived from instances where there were technical glitches due to long lasting experiment in a day and overload of the humanoid robot. Especially there were several incidents where voice recognition that the robot did not understand the participants' instruction at once and had to repeat the instruction or response several times. For example, Participant #20 mentioned, "Voice recognition didn't work well on the treadmill, which made communication difficult at first." However, this was solved when there were enough block between experiment sessions for the robot to cool down and if this issue occurred there were enough time to debug the system.

The humanoid aspect of the robot were also led to different responses among participants, where many participants thought the robot is friendly and cute, but some others thought they were somewhat 'creepy' because of its appearance. This is a constant phenomenon in human-robot interaction where robots with more human-like appearance lead to perceived discomfort ('uncanny valley effect'[45]).

Participants generally agreed that the robot coach effectively motivated those less inclined to exercise regularly. Participant #6 reported, "For those who don't exercise much, having a robot alongside was a good motivator."

The qualitative analysis revealed that humanoid robots can be effective coaches, especially for individuals less inclined towards regular exercise. While regular exercisers did not alter their routines significantly, less frequent exercisers showed a notable increase in intensity. The robot's presence and real-time feedback were highly valued, although technical improvements are needed.

These findings suggest promising directions for future research and practical applications in fitness regimes using humanoid robots.

5. Discussion

From our study, we found that in the context of fitness coaching, robot's factual feedback actually increase exercise intensity more than empathic feedback.

The concrete and directive nature of factual feedback significantly impacted participants' decisions to walk faster compared to emotionally supportive feedback. Concise feedback helps people process information more quickly, particularly when they have limited cognitive capacity to do so in real-time, such as when walking on a treadmill and listening to instructions simultaneously[46]. Delivering only necessary information can help people perceive more control over the situation, reinforcing their motivation to exercise intensively[47].

However, it is possible that the initial stages of the exercise session, including ice-breaking and stretching, made participants feel more comfortable. According to social presence theory, people treat computers or virtual characters as social actors and may see the robot as a companion. This initial comfort and perceived companionship likely provided emotional support, which may have enhanced the effectiveness of the factual feedback. This pattern is consistent with our qualitative evaluation, where participants regarded the robot coach as more motivating and enjoyable.

The discrepancy in the feedback effect may be due to differences in the robotic interface and feedback delivery methods. Using a humanoid robot with a more human-like appearance and movements that mimic humans may have given the impression that it could empathize with humans, leading to increased acceptance[4,5,18]. Future research could explore the use of avatars in virtual reality or non-humanoid robots.

Additionally, individual differences in perceiving feedback may yield different results. In this study, we found differences in perceived usability and trust in the robot coach's instruction depending on the expertise level of the user (regular exercisers vs. non-regular exercisers). More user-specific factors such as age, personality, and other demographic factors should be explored.

Our findings show that an appropriate coaching style increases exercise intensity, as evidenced by faster walking speeds and elevated heart rates. The changes in heart rate and walking speed highlight the significance of tailoring feedback to individual needs. Future research should consider personalized feedback targeting specific levels of experience or personal needs, such as different expertise levels and exercise preferences (e.g., cardio vs. strength training[41])

Implementing personalized feedbacks will require different experimental design from our one-time 30-minute exercise session, which was more suitable for laypeople who exercise less. Instead, shorter and regular check-ups in a long-term study are appropriate for these specific needs and patients requiring regular and repetitive sessions (e.g., stroke rehabilitation[8,9]). Robotic coaching can be valuable in enhancing exercise motivation and performance across diverse user groups, such as children[48] and older adults[7].

These findings have important implications for the system design targeting users' physical health. Our results show that using a robot coach can effectively increase exercise motivation, especially for people who exercise less frequently. This approach would be particularly helpful if implemented in programs targeting people who work less frequently or people who have sedentary lifestyles[6]).

6. Conclusions

Our study focused on two distinct styles of encouragement provided by the robot coach: factual and empathetic feedback. We found that feedback from robot coaches focused on providing information rather than emotional support resulted in immediate behavioral changes such as increased walking speed and heart rate, and improved exercise efficiency. Our qualitative analysis shows that exercising with a robot coach resulted in overall satisfactory and positive experiences, which could potentially affect long-term exercise habit formation[49,50].

In short, factual and concise feedback have an immediate effect in increasing exercise intensity. The robot's social presence and emotionally supportive manner will make the overall experience enjoyable, increasing long-term motivation, especially for non-regular exercisers.

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Appendix I

	Item	Mean	SD
Acceptance	1. I like to use robot for exercise	3.58	0.92
	2. I'm willing to use the robot again for exercise	3.87	0.92
	3. The smart watch was uncomfortable *	4.4*	0.78
	4. The robot showed appropriate emotional expressions	3.42	0.81
Perceived Usefulness	5. The exercise that I did through the robot's guidance was good for my health	4.27	0.62
	6. The robot was not helpful for exercise *	4.11*	0.68
	7. The robot clearly showed each posture	3.09	0.82
	8. The robot was difficult to use *	4.07*	0.78
	9. The robot motivated me to exercise	4.00	0.80
	10. I don't trust the robot's advice *	3.87	0.79
Robot Appearance	11. The robot moves too fast for me to follow *	4.83*	0.72
	12. The robot has a clear voice	3.56	1.01
	13. I didn't quite understand the robot's instructions *	4.09*	0.82
	14. I think the size of the robot is appropriate for exercise	3.60	0.84

*: Reverse-Coded Score (6 - Original Score)